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Learning from heterogeneous structural MRI via collaborative domain adaptation for late-Life depression assessment

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ABSTRACT

Accurate identification of late-life depression (LLD) using structural brain MRI is essential for monitoring disease progression and facilitating timely intervention. However, existing learning-based approaches for LLD detection are often constrained by limited sample sizes (e.g., tens), which pose significant challenges for reliable model training and generalization. Although incorporating auxiliary datasets can expand the training set, substantial domain heterogeneity, such as differences in imaging protocols, scanner hardware, and population demographics, often undermines cross-domain transferability. To address this issue, we propose a Collaborative Domain Adaptation (CDA) framework for LLD detection using T1-weighted MRIs. The CDA leverages a dual-branch architecture integrating a Vision Transformer (ViT) and a Convolutional Neural Network (CNN) to exploit their complementary strengths in capturing global anatomical topology and local texture details. The CDA framework consists of three stages: (a) *supervised training* on labeled source data, (b) *self-supervised target feature adaptation* to refine decision boundaries, and (c) *collaborative training* on unlabeled target data. The collaborative stage employs a reliability-aware JSD-based dual-constraint mechanism to filter high-quality pseudo-labels, encouraging robust prediction consistency without target supervision. Extensive experiments on two multi-site benchmarks demonstrate that CDA consistently outperforms state-of-the-art unsupervised domain adaptation methods, highlighting its superior generalization ability for real applications.

1. Introduction

Late-life depression (LLD) is a prevalent and debilitating neuropsychiatric disorder that affects a substantial proportion of the aging population. It poses significant challenges to cognitive health, quality of life, and overall mortality rates (Alexopoulos, 2005; Blazer, 2003). Early and accurate identification of LLD is essential to facilitate timely intervention and mitigate the risk of progression to more severe cognitive impairment. Neuroimaging, particularly structural magnetic resonance imaging (MRI), has emerged as a valuable solution for exploring the neurobiological basis of LLD. Prior MRI-based studies have reported that LLD is associated with widespread structural abnormalities, particularly in regions such as the prefrontal cortex, hippocampus, and major white matter tracts (Krishnan et al., 1997; Mortimer et al., 2013; Sawyer et al., 2012).

Learning-based approaches, including both traditional and deep models, have demonstrated potential in MRI biomarker identification

and automated classification of LLD (Herrmann et al., 2008; Sexton et al., 2013). Unfortunately, these models rely on large-scale training data to achieve robust performance (Eitel et al., 2021; Jónsson et al., 2019; Knoll et al., 2020; Mårtensson et al., 2020; Wen et al., 2020), whereas LLD-related MRI datasets are typically limited in size. While aggregating multiple neuroimaging datasets can increase sample size and improve model generalization, this approach often introduces domain shift (Ganin & Lempitsky, 2015; Mårtensson et al., 2020; Sarafraz et al., 2024), which refers to statistical differences caused by heterogeneous scanner types, imaging protocols, and subject populations.

Domain adaptation techniques have attracted increasing attention in these data-constrained medical imaging scenarios, aiming to reduce the discrepancy between source and target distributions (Guan & Liu, 2023). Classical methods include adversarial training (Ganin & Lempitsky, 2015), maximum mean discrepancy (MMD) minimization (Long et al., 2015), and self-supervised consistency regularization (Berthelot et al., 2019; French et al., 2018). These techniques have demonstrated

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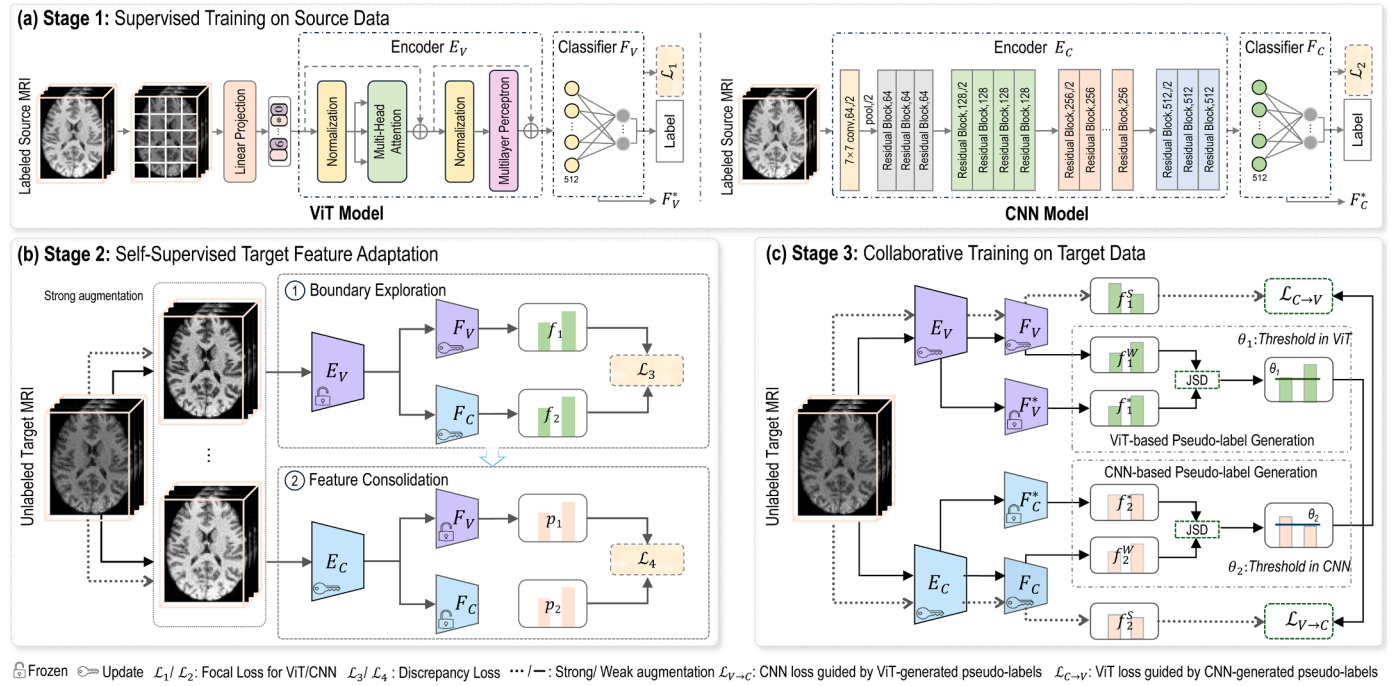


Fig. 1. Illustration of collaborative domain adaptation (CDA), consisting of (a) supervised training on labeled source data, (b) self-supervised target feature adaptation between ViT and CNN branches to refine decision boundaries, and (c) collaborative training using pseudo-labeled and augmented target samples to enhance prediction consistency and domain generalization. Legend and Notations: Solid arrows indicate the forward data flow (e.g., strongly augmented samples), while dashed arrows represent the flow of supervisory signals or guidance from weakly augmented samples. The lock icon denotes frozen parameters that are not updated during the specific phase. The markers θ_1 and θ_2 explicitly define the confidence thresholds used to filter high-quality pseudo-labels for the ViT and CNN branches, respectively. The ViT branch consists of an encoder E_V and a classifier F_V , while the CNN branch comprises an encoder E_C and a classifier F_C . JSD: Jensen-Shannon divergence.

promising results in medical imaging tasks such as brain tumor segmentation (Dou et al., 2018), Alzheimer's disease classification (Agarwal et al., 2021; Pan et al., 2021), and schizophrenia prediction (Liu et al., 2023). However, current domain adaptation approaches face two key limitations. First, they often treat the model as a fixed, uniform system and focus solely on aligning feature distributions across domains, often overlooking the complementary nature of different model architectures or representations (Pei et al., 2018; Xu et al., 2020). Second, they often struggle to achieve robust performance when the target domain has extremely limited data, a common issue in the context of LLD (Perone et al., 2019; Zhao et al., 2021).

To this end, we propose a novel Collaborative Domain Adaptation (CDA) framework for MRI-based identification of LLD. Our framework integrates a Vision Transformer (ViT) (Dosovitskiy et al., 2021) to model global anatomical features and a Convolutional Neural Network (CNN) (He et al., 2016) to capture local structural patterns from MRI. The complementary nature of these architectures enables richer representation learning. As illustrated in Fig. 1, the CDA framework comprises three stages: (a) supervised training of ViT and CNN on labeled source data, (b) self-supervised target feature adaptation, and (c) collaborative training on unlabeled target data. The first stage uses supervised learning on the source domain to ensure that ViT and CNN branches have basic discriminative capabilities. During the self-supervised target adaptation stage, the two models are fine-tuned through feature alignment to minimize the discrepancy between the classifier outputs of the two branches, thereby refining the decision boundaries. The subsequent collaborative training stage utilizes pseudo-labeled and augmented target-domain MRIs to enforce prediction consistency under both strong and weak perturbations, enhancing domain robustness and generalization. Distinct from recent collaborative strategies (Ngo et al., 2021; Yang et al., 2022) designed for Semi-Supervised Domain Adaptation (SSDA) that rely on partial target annotations to anchor alignment, our framework oper-

ates in a strictly **Unsupervised (UDA)** manner. It is specifically designed to mitigate the risk of mode collapse in the absence of target supervision through collaborative consensus. This collaborative strategy effectively addresses both the data scarcity and domain shift problems by leveraging source knowledge while promoting generalization to the target domain. Experiments on multi-site T1-weighted MRI data related to late-life depression demonstrate that CDA significantly outperforms several state-of-the-art domain adaptation baselines across multiple evaluation metrics. The source code has been released to the public through GitHub¹.

The contributions of this work are summarized as follows.

- We propose a Collaborative Domain Adaptation (CDA) framework tailored for late-life depression identification. To our knowledge, this is one of the first efforts to integrate CNN and ViT in a dual-branch cross-supervision architecture, effectively leveraging their complementary strengths (local texture vs. global topology) to address heterogeneous domain shifts.
- We introduce a reliability-aware collaborative training strategy that employs a JSD-based dual-constraint mechanism. This allows the model to safely learn from unlabeled target data by filtering noise, thereby mitigating the risk of mode collapse in strictly unsupervised settings.
- We conduct extensive validations on two multi-site benchmarks. The results demonstrate that CDA consistently outperforms state-of-the-art methods, verifying its robustness for real-world clinical deployment.

The remainder of this paper is organized as follows. Section 2 reviews the relevant studies. Section 3 introduces materials and data

¹ <https://github.com/yzgao2017/CDA>

pre-processing. Section 4 provides a comprehensive description of the proposed method. Section 5 introduces experimental setup, competing methods, and experimental results. Section 6 presents an ablation study, analyzes the influences of key CDA components, and discusses the limitations of the current work. This paper is concluded in Section 7.

2. Related work

2.1. Late-Life depression progression analysis

Late-life depression (LLD) is a prevalent but often underdiagnosed neuropsychiatric disorder affecting a significant proportion of the elderly population. Growing evidence links LLD to accelerated cognitive decline and an increased risk of progression to dementia (Butters et al., 2008; Mast et al., 2008). To better understand its neurobiological underpinnings, structural magnetic resonance imaging (MRI) has been widely used to investigate neuroanatomical changes associated with LLD. Consistent alterations have been reported in brain regions including the prefrontal cortex, hippocampus, amygdala, and anterior cingulate cortex (Bora et al., 2012; Lai, 2013). Recent studies have increasingly focused on characterizing the progression of LLD and its potential biomarkers using structural MRI (Alexopoulos et al., 2009; Gunning et al., 2009; Koenig & Butters, 2014). However, many of these investigations rely on small, homogeneous cohorts or data collected at a single site, which limits statistical power and hinders the generalizability of the findings (Benjamin & Steffens, 2011; Kanellopoulos et al., 2011; Taylor et al., 2013).

Given the scarcity of large-scale, multi-site datasets for LLD, recent efforts have focused on pooling data from heterogeneous sources or conducting cross-cohort analyses to increase sample size and diversity (Tan et al., 2023; Wen et al., 2021). While such strategies can enhance the representativeness of findings, they also introduce substantial *domain shifts* due to differences in scanner hardware, imaging protocols, clinical labeling standards, and population characteristics. These variations significantly affect model performance and reproducibility, particularly in neuroimaging-based diagnostic tasks, where models trained on one dataset often struggle to generalize to others without proper domain adaptation strategies (Ghafoorian et al., 2017; Wachinger et al., 2021; Wolleb et al., 2022). These challenges highlight the urgent need for robust, domain-adaptive learning approaches capable of addressing cross-site heterogeneity in 3D structural MRI for reliable LLD diagnosis.

2.2. Domain adaptation for MRI analysis

Domain adaptation (DA) has become an essential research direction for mitigating domain shift between training and testing distributions, particularly in medical imaging, where variability across scanners, institutions, and acquisition protocols is inevitable. Among various DA paradigms, unsupervised domain adaptation (UDA) has gained increasing attention for its potential to improve generalization in real-world clinical deployments (Jiang et al., 2022; Tzeng et al., 2017), which assumes access to labeled source data and unlabeled target data.

Common strategies in UDA include adversarial learning (Kamnitsas et al., 2017; Tzeng et al., 2017), distribution alignment via maximum mean discrepancy (MMD) (Long et al., 2015), and entropy minimization (Vu et al., 2019). More recently, a high-order graph learning approach has been proposed to model complex data distributions and improve context-aware adaptation (Zhu et al., 2025). In neuroimaging, these techniques have been applied to tasks such as tumor segmentation (Kamnitsas et al., 2017) and Alzheimer's staging (Guan et al., 2021). Most existing DA models rely on CNN-based architectures due to their local receptive field and inductive bias towards translational invariance. However, CNNs often fall short in capturing long-range dependencies and holistic anatomical patterns, especially in high-resolution 3D MRI volumes.

Table 1

Demographic information and category labels of the subjects in cognitive impairment-related studies from different data domains in the article. The values are denoted as “mean $\pm$ standard deviation”. F/M: Female/Male, MMSE: Mini-Mental State Examination.

Domain	Dataset	Category	Gender (F/M)	Age	MMSE
Target	NBOLD	CN-N	27/7	71.6 $\pm$ 7.1	29.1 $\pm$ 1.5
		CN-D	50/16	70.0 $\pm$ 6.9	29.5 $\pm$ 0.8
		CIND	6/3	74.9 $\pm$ 4.3	29.2 $\pm$ 1.2
		AD	3/5	76.5 $\pm$ 7.5	28.5 $\pm$ 1.7
Source	NCODE	CN-N	33/23	67.5 $\pm$ 5.0	29.1 $\pm$ 1.0
		CN-D	69/41	66.1 $\pm$ 5.7	28.5 $\pm$ 1.4
		CIND	5/5	71.9 $\pm$ 6.3	27.4 $\pm$ 2.2
		AD	5/3	72.0 $\pm$ 5.6	24.8 $\pm$ 5.4

In contrast, Vision Transformers (ViTs) have emerged as powerful alternatives, offering the ability to model global anatomical context through self-attention mechanisms. Recent models such as TransUNet (Chen et al., 2021) and UNETR (Hatamizadeh et al., 2022) have demonstrated state-of-the-art performance in medical image segmentation and classification. Nevertheless, most existing UDA approaches rely exclusively on either CNN or ViT backbones (Huang et al., 2020; Liu et al., 2021), failing to fully exploit the complementary strengths of both. Our work addresses this gap by introducing a hybrid domain adaptation framework that integrates ViT and CNN in a collaborative dual-branch architecture. This design leverages ViT's ability to capture global context and CNN's strength in modeling fine-grained local features, resulting in a more comprehensive feature representation.

2.3. Collaborative learning in domain adaptation

Collaborative learning strategies have been proposed to improve DA model robustness under limited target supervision. In particular, co-training (Blum & Mitchell, 1998) and mutual pseudo-labeling approaches (Shu et al., 2018; Tang et al., 2020; You et al., 2019) iteratively refine predictions across multiple views or learners, enhancing model stability in UDA settings. These techniques are especially valuable in medical imaging tasks, where obtaining ground-truth labels in the target domain is costly or infeasible. More recently, hybrid ViT-CNN architectures have been explored to capture complementary structural priors in complex imaging data (Dong et al., 2025; Xu et al., 2023). A recent study (Ngo et al., 2024) introduced an explicit class-specific boundaries (ECB) learning framework, in which the ViT branch maximizes classifier discrepancy to explore class boundaries, while the CNN branch minimizes this discrepancy to cluster features within each class. This asymmetric design enhances decision margin clarity and enables more accurate pseudo-label propagation.

However, our work differs from these recent hybrid approaches (Ngo et al., 2024) and collaborative strategies (Ngo et al., 2021; Yang et al., 2022) in two fundamental aspects tailored for heterogeneous medical imaging. First, regarding pseudo-label reliability, methods like ECB typically filter pseudo-labels using a single confidence threshold (e.g., $P > \tau$). In heterogeneous MRI adaptation, where scanner artifacts often cause probability miscalibration, this can lead to error accumulation. In contrast, our CDA introduces a reliability-aware JSD mechanism that enforces a “dual-constraint”: samples are selected only if they satisfy both high confidence and high cross-view consistency, acting as a stability filter against domain noise. Second, regarding supervision settings, unlike collaborative works that rely on semi-supervised domain adaptation (SSDA) with labeled target data (Zhang et al., 2024), CDA operates in a strictly unsupervised manner. Our framework mitigates the risk of mode collapse through collaborative consensus, without relying on ground-truth anchors in the target domain.

Table 2
Diagnostic category and description of the studied participants..

Category	Description
CN-N	Cognitively Normal
CN-D	Depressed, Cognitively Normal
CIND	Cognitive Impairment, No Dementia; Cognitive Impairment, due to Vascular Disease
AD	Probable AD, Possible AD, Subsyndromal AD; Dementia of undetermined etiology

3. Materials

3.1. Studied subjects

This study uses multi-site T1-weighted MRI data related to late-life depression from two imaging cohorts: 1) the Neurobiology of Late-life Depression (NBOLD) database (Steffens et al., 2017), and 2) the Neurocognitive Outcomes of Depression in the Elderly (NCODE) study (Steffens et al., 2004). The demographic and clinical characteristics of these participants are reported in Table 1.

NBOLD. The NBOLD study collected T1-weighted MRI scans between 2013 and 2017 at the Olin Neuropsychiatric Research Center. The scans were acquired using a Skyra 3T MRI scanner with MPRAGE protocol (TR/TE = 2200/2.88 ms, voxel size = $0.8 \times 0.8 \times 0.8$ mm). All participants were older adults assessed for late-life depression (LLD) and followed up for cognitive outcomes. Participants were enrolled at baseline with cognitive normality and monitored for five years. Diagnoses were assigned based on five-year follow-up outcomes and clinical consensus, resulting in four categories: 1) never-depressed, cognitively normal (CN-N); 2) depressed, cognitively normal (CN-D); 3) cognitively impaired but without dementia (CIND); and 4) Alzheimer's disease (AD), as presented in Table 2. All diagnoses were confirmed by experienced geriatric psychiatrists using standardized protocols and clinical scales. Considering the limited sample size of CIND and AD, we combined these two groups into the cognitive impairment (CI) category. This dataset was used as the *target domain* in this work, including 34 CN-N subjects, 66 CN-D subjects, and 17 CI subjects (CIND: 9, AD: 8).

NCODE. The NCODE includes 3T T1-weighted MRI scans collected between 1995 and 2013 using a Siemens Trio MRI scanner. These scans were acquired using 3D TURBOFLASH sequence (TR/TE = 22/7 ms, voxel size = $1 \times 1 \times 1$ mm). These scans provide detailed brain structural information relevant to both depression and cognitive decline analysis. As in NBOLD, all participants were cognitively normal at baseline and were followed for five years. Their diagnostic outcomes at the five-year follow-up were used for classification. Subjects were categorized into three diagnostic groups: CN-N, CN-D, and CI (with CIND and AD). This dataset was used as the *source domain* in this work, consisting of 56 CN-N subjects, 110 CN-D subjects, and 18 CI subjects (CIND: 10, AD: 8).

3.2. Data pre-Processing

All MRI scans underwent a standard preprocessing pipeline to ensure consistency across datasets. First, N4 bias field correction was applied to address intensity inhomogeneities (Tustison et al., 2010). Then, images were linearly and non-linearly registered to the MNI space using the Advanced Normalization Tools (ANTs) (Avants et al., 2011). Skull stripping was performed using a robust brain extraction method. All images were resampled to 1 mm isotropic resolution and normalized in shape to the Automated Anatomical Labeling 3 (AAL3) atlas (Rolls et al., 2020), which includes 170 regions of interest (ROIs). Structural brain features such as cortical thickness, surface area, and gray matter volume were extracted using FreeSurfer (Fischl, 2012). Histogram matching and z-score normalization were performed using TorchIO (Pérez-García et al., 2021) to standardize intensity distributions across sites. The final images were

standardized to $181 \times 217 \times 181$ voxels in the Montreal Neurological Institute (MNI) space, ensuring spatial correspondence across all datasets.

4. Methodology

4.0.0.1. Problem Formulation. The goal is to identify patients with LLD from cognitively normal subjects in a target domain with limited data by leveraging knowledge from relatively large-scale source data. We have a labeled source domain $D_S = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$, where each 3D MRI $x_i^s \in \mathbb{R}^{H \times W \times D}$ corresponds to a ground-truth category label y_i^s , and an unlabeled target domain $D_T = \{x_i^t\}_{i=1}^{N_t}$, where N_s is the number of source samples and N_t is the number of target samples. As illustrated in Fig. 1, the proposed CDA has two branches: 1) ViT for global MRI feature learning, and 2) CNN for local MRI feature learning. This dual-branch design is motivated by the fact that ViT can effectively capture long-range dependencies and model comprehensive global representations, while CNN is good at capturing local spatial hierarchies for robust feature learning. To facilitate knowledge transfer from source to target domain, we develop three components in CDA, including 1) *supervised training* on labeled source data, 2) *self-supervised target feature adaptation*, and 3) *collaborative training* of ViT and CNN on unlabeled target data. To improve cross-domain generalization, we employ data augmentation in CDA, applying weak and strong augmentations (Cardoso et al., 2022) to enhance their feature consistency.

4.1. Supervised training on labeled source data

As shown in Fig. 1 (a), each input MRI image is simultaneously processed by two distinct branches: a ViT and a CNN, each comprising an encoder and a classifier. The ViT branch first divides the 3D MRI into non-overlapping patches, which are then embedded into tokens and pre-processed by an encoder E_V composed of multiple self-attention layers. The resulting global feature representation is fed into a classifier F_V for category prediction. In parallel, the CNN branch extracts hierarchical local features through an encoder E_C consisting of stacked convolutional layers and residual blocks. These features are then input to the CNN classifier F_C , yielding the corresponding class logits. To ensure effective feature learning, both branches are independently trained on labeled source domain samples. Given the potential class imbalance in the source data, we employ the focal loss (Lin et al., 2017) to increase the model's sensitivity to hard or minority class examples. The optimization objectives for the two branches are defined as:

$$\begin{aligned} \mathcal{L}_1 &= \frac{1}{N_s} \sum_{i=1}^{N_s} \mathcal{L}_{focal}(y_i^s, \sigma(F_V(E_V(x_i^s))))), \\ \mathcal{L}_2 &= \frac{1}{N_s} \sum_{i=1}^{N_s} \mathcal{L}_{focal}(y_i^s, \sigma(F_C(E_C(x_i^s))))), \end{aligned} \quad (1)$$

where $\mathcal{L}_{focal}(\cdot)$ is the focal loss, $\sigma(F_V(E_V(x_i^s)))$ and $\sigma(F_C(E_C(x_i^s)))$ denote the predicted probability scores from the ViT and CNN branches, respectively, and y_i^s is the ground-truth category label of the source sample x_i^s . Here, the softmax function $\sigma(\cdot)$ is used to convert logits into normalized class probabilities. The generated classifiers (i.e., F_V^* and F_C^*) trained on labeled source data will be utilized in the third stage of the CDA framework.

4.2. Self-Supervised target feature adaptation

In this work, we aim to bridge the domain gap between the source and target distributions by aligning their feature representations in a self-supervised manner. Specifically, we leverage the complementary properties of ViT and CNN branches: the ViT encoder defines sharper and more discriminative class boundaries, while the CNN encoder adjusts its representations to align with those boundaries. This module comprises two sequential phases, as illustrated in Fig. 1 (b): (1) *Boundary Exploration*, and (2) *Feature Consolidation*.

1) Boundary Exploration: In this phase, the ViT encoder E_V is frozen, and the classifiers F_V and F_C are fine-tuned to maximize their predictive divergence on unlabeled target samples. This encourages the two classifiers to expand their decision boundaries, facilitating the identification of target features that deviate from the support of source distribution. To quantify disagreement, we define a discrepancy loss between class probability outputs $\mathbf{a}$ and $\mathbf{b}$ of the two classifiers as:

$$D\mathcal{L}(\mathbf{a}, \mathbf{b}) = \frac{1}{K} \sum_{k=1}^K |a_k - b_k|, \quad (2)$$

where K is the number of classes, and a_k and b_k denote the predicted probabilities from F_V and F_C for the k -th class. Specifically, we fix the encoder E_V (trained in Stage 1) and update the classifiers F_V and F_C to maximize their prediction difference on target data. The objective function is defined as:

$$\begin{aligned} \mathcal{L}_3 = & -\frac{1}{N_t} \sum_{i=1}^{N_t} D\mathcal{L}(f_1, f_2) \\ & + \frac{1}{N_s} \sum_{i=1}^{N_s} \mathcal{L}_{focal}(y_i^s, \sigma(F_V(E_V(x_i^s)))) \\ & + \frac{1}{N_s} \sum_{i=1}^{N_s} \mathcal{L}_{focal}(y_i^s, \sigma(F_C(E_C(x_i^s)))) \end{aligned} \quad (3)$$

where $f_1 = \sigma(F_V(E_V(x_i^t)))$ and $f_2 = \sigma(F_C(E_C(x_i^t)))$ are the softmax-normalized outputs of the classifiers for the unlabeled target sample x_i^t . Note that both encoders E_V and E_C include a projection layer to map features to the same dimension (i.e., 512) before the classifiers, ensuring architectural compatibility for this cross-branch operation. Minimizing $\mathcal{L}_3$ encourages the classifiers to disagree on ambiguous target samples, thereby delineating class boundaries. We also input labeled source data to ViT and CNN for supervised training by minimizing the focal losses (i.e., the last two focal loss terms in Eq. (3)), where their encoders (i.e., E_V and E_C) are kept frozen and only the classifiers (i.e., F_V and F_C) are updated. This enables the two classifiers to accurately distinguish between different categories without being misled by challenging target samples (Saito et al., 2018). During training, we maintain a balanced number of source and target samples in each mini-batch to facilitate effective domain adaptation.

2) Feature Consolidation: To improve feature consistency across encoders, we refine the CNN encoder E_C to better align its extracted features with the class boundaries defined by the ViT encoder E_V , particularly in regions of high uncertainty identified via classifier divergence. In this phase, the classifiers F_V and F_C are frozen, and E_C is optimized to minimize the discrepancy between their predictions for target data:

$$\mathcal{L}_4 = \frac{1}{N_t} \sum_{i=1}^{N_t} D\mathcal{L}(p_1, p_2), \quad (4)$$

where $p_1 = \sigma(F_V(E_C(x_i^t)))$ and $p_2 = \sigma(F_C(E_C(x_i^t)))$ denote the outputs of the two classifiers when applied to features extracted by the CNN encoder E_C . This procedure encourages the CNN to produce representations that are consistent with the previously established decision boundaries, thereby consolidating the feature space. Aligning features in this manner promotes classification robustness and generalization, particularly in the absence of labeled target samples.

4.3. Collaborative training on unlabeled target data

This phase aims to mitigate the discrepancy between the two network branches by enabling mutual reinforcement, leveraging their complementary strengths to generate reliable pseudo-labels for the unlabeled target samples. Notably, this approach enhances the CNN branch by exploiting the ViT's superior capacity for modeling complex global patterns. To facilitate effective cross-branch knowledge transfer, pseudo-labels are generated by the ViT from weakly augmented target samples ($x^{t,W}$) and then employed to supervise the CNN branch on strongly augmented counterparts ($x^{t,S}$). Similarly, the CNN also guides the ViT under the same paradigm, as illustrated in Fig. 1 (c). Strong augmentations are realized through random large-scale affine transformations and elastic deformations, while weak augmentations include

random flipping and mild affine adjustments (Cardoso et al., 2022). To enhance the fidelity of pseudo-labels, a Jensen-Shannon divergence (JSD)-based optimization mechanism (Menéndez et al., 1997) is incorporated.

1) Pseudo-Label Generation: For each unlabeled target instance, feature extraction is performed by the encoders E_V and E_C , followed by classification via F_V and F_C to obtain pseudo-label predictions. The preliminary classifiers F_V^* and F_C^* , trained during the supervised stage, provide a reference for consistency evaluation on similarly augmented samples. The classification outputs on a set of weakly augmented inputs $x^{t,W}$ are denoted as: (a) $q_1^* = F_V^*(E_V(x^{t,W}))$ and $q_1 = F_V(E_V(x^{t,W}))$ for the ViT branch, where F_V is the current classifier; (b) $q_2^* = F_C^*(E_C(x^{t,W}))$ and $q_2 = F_C(E_C(x^{t,W}))$ for the CNN branch, where F_C is the current classifier. The Jensen-Shannon divergence (JSD) quantifies the consistency between preliminary and current classifier outputs within each branch, defined as:

$$\text{JSD}(P_1, P_2) = \frac{1}{2} D_{KL}(P_1 \| M) + \frac{1}{2} D_{KL}(P_2 \| M), \quad (5)$$

where $M = \frac{1}{2}(P_1 + P_2)$ denotes the mixture distribution and D_{KL} represents the Kullback-Leibler divergence (Kullback & Leibler, 1951). Consistency measures $\text{JSD}(q_1^*, q_1)$ and $\text{JSD}(q_2^*, q_2)$ are computed for ViT and CNN respectively. When the divergence falls below a predefined threshold $\tau = 0.1$, predictions are deemed sufficiently consistent, thus qualifying the corresponding samples for high-confidence pseudo-label generation. A target sample is selected for pseudo-labeling only if it simultaneously satisfies the JSD-based consistency constraint for the corresponding branch (i.e., $\text{JSD}(q^*, q) \leq \tau$) and the confidence threshold applied in the subsequent collaborative training stage. For such target samples within a batch, pseudo-labels are computed as:

$$\hat{y}_V^t = \frac{1}{2}(f_1^* + f_1^W), \quad \hat{y}_C^t = \frac{1}{2}(f_2^* + f_2^W), \quad (6)$$

where $f_1^* = \sigma(q_1^*)$, $f_1^W = \sigma(q_1)$, $f_2^* = \sigma(q_2^*)$, and $f_2^W = \sigma(q_2)$, and $\sigma(\cdot)$ denotes the softmax function.

2) Collaborative Training: To foster cross-branch learning, we employ pseudo-labels $\hat{y}_V^t$ generated by ViT for weakly augmented samples to supervise the CNN branch on strongly augmented inputs $x^{t,S}$ by minimizing:

$$\mathcal{L}_{V \rightarrow C} = CE(F_C(E_C(x^{t,S})), \hat{y}_V^t) \cdot \mathbf{1}(\max(\hat{y}_V^t) > \theta_1), \quad (7)$$

where $\mathbf{1}(\cdot)$ is the indicator function selecting only those samples with confidence exceeding threshold θ_1 , and $CE(\cdot)$ denotes the cross-entropy loss. Similarly, the CNN-generated pseudo-labels $\hat{y}_C^t$ supervise the ViT branch through:

$$\mathcal{L}_{C \rightarrow V} = CE(F_V(E_V(x^{t,S})), \hat{y}_C^t) \cdot \mathbf{1}(\max(\hat{y}_C^t) > \theta_2), \quad (8)$$

where θ_2 serves as the confidence threshold for CNN. These bidirectional supervisory signals promote effective knowledge exchange between the ViT and CNN branches, enhancing their respective capabilities in capturing global contextual information and local spatial hierarchies, encouraging robust MRI feature learning. During inference, we use the trained CNN model for prediction, with more discussions presented in Section 6.5.

4.4. Implementation

In this work, we use a lightweight ViT architecture (Touvron et al., 2021) and a ResNet-34 CNN model from the MONAI framework (Cardoso et al., 2022) in the proposed CDA framework. To leverage large-scale auxiliary data, the ViT encoder E_V is pretrained using a self-supervised Masked Autoencoder (MAE) approach on unlabeled brain MRI datasets (i.e., IXI, OASIS-3, and BRATS) (Kunanbayev et al., 2024). The CNN encoder E_C is pretrained on 9544 T1-weighted MRIs from ADNI (Jack et al., 2008) via an autoencoder-based unsupervised image reconstruction approach (Zhang et al., 2025). Each classifier (F_V for ViT and F_C for CNN) consists of two fully connected layers followed by

a softmax activation function. Model optimization employs Stochastic Gradient Descent (SGD) (Bottou, 2012) with momentum 0.9 to accelerate convergence and escape local minima. A weight decay of 0.0005 provides l_2 regularization to improve generalization. Learning rates for both encoders are set to 1×10^{-4} and 5×10^{-4} , respectively, appropriate for fine-tuning pretrained models. The batch size is set to 4, balancing GPU memory constraints and gradient stability. Pseudo-labeling in collaborative training applies confidence thresholds specific to each branch: $\theta_1 = 0.5$ for ViT-generated labels and a more stringent $\theta_2 = 0.8$ for CNN. These empirically chosen thresholds ensure that only high-confidence pseudo-labels contribute to model training.

5. Experiments

5.1. Experimental setup

Classification Task. We carried out two classification tasks: (1) binary classification between CN-D and CN-N on the target domain (NBOLD), and (2) three-category classification among CI, CN-D, and CN-N on the target domain (NBOLD). For binary classification, the mean and standard deviation of the five evaluation metrics were computed across the five cross-validation folds. For three-category classification, we employed the widely used one-vs-rest strategy (Aly, 2005). Predictions across all cross-validation folds were aggregated to calculate the overall accuracy (ACC) and sensitivity (SEN), along with per-class ACC and SEN scores.

Evaluation Metric. To comprehensively assess model performance, we employed several widely used evaluation metrics, including area under the receiver operating characteristic curve (AUC), accuracy (ACC), sensitivity (SEN), specificity (SPE), and F1-score (F1). All experiments were conducted under a five-fold cross-validation (CV) strategy for the target data. Specifically, target subjects within each class were randomly partitioned into five (approximately equal-sized) subsets. In each iteration, one subset was held out as the test set, while the remaining four subsets were used for training. This CV procedure was repeated five times with different random seeds to further minimize variance, and to reduce potential bias introduced by sample partitioning.

Competing Methods. We compare the proposed CDA with a comprehensive set of domain adaptation methods that are commonly used in heterogeneous structural MRI analysis, covering both traditional feature-based approaches and recent deep learning-based methods. Specifically, three classical domain adaptation methods are evaluated using 1,373-dimensional handcrafted features extracted by FreeSurfer, while ten state-of-the-art deep domain adaptation methods are evaluated using 3D MRI scans as input. For all competing methods, we follow the default settings reported in the original papers and adopt comparable backbone architectures and hyperparameters to ensure a fair comparison. To ensure that performance gains are not solely attributed to the pre-training scale, all CNN-based deep learning baselines were initialized with the same pre-trained weights (via autoencoder pre-training on $\sim 9.5k$ ADNI MRIs) as used in our CNN branch. For ViT-based methods, including ViT-ADA (Li & Wu, 2024), the same MAE-based pre-training strategy was employed wherever architecturally applicable.

Traditional domain adaptation methods. Transfer Component Analysis (TCA) (Pan et al., 2010), Joint Distribution Adaptation (JDA) (Long et al., 2013), and Scatter Component Analysis (SCA) (Ghifary et al., 2016) are representative kernel-based methods that aim to reduce domain discrepancies by learning shared subspaces across source and target domains. These methods have been widely used as classical baselines in cross-site structural MRI studies.

Deep distribution alignment methods. Deep Adaptation Network (DAN) (Long et al., 2015), DeepCORAL (Sun & Saenko, 2016), and Deep Subdomain Adaptation Network (DSAN) (Zhu et al., 2020) minimize statistical discrepancies (e.g., MMD, covariance, or local distribution matching) to align feature representations.

Adversarial learning strategies are employed by Domain-Adversarial Neural Network (DANN) (Ganin & Lempitsky, 2015), Adversarial Discriminative Domain Adaptation (ADDA) (Tzeng et al., 2017), and Dynamic Adversarial Adaptation Network (DAAN) (Yu et al., 2019) to encourage domain invariance via global or class-conditional objectives.

Recent MRI-oriented and prediction-based methods. Batch Nuclear-norm Maximization (BNM) (Cui et al., 2020) enhances prediction discriminability on unlabeled targets, while Source Hypothesis Transfer++ (SHOT++) (Liang et al., 2021) enables source-free adaptation, particularly relevant for privacy-preserving medical imaging. Furthermore, Vision Transformer-based Adversarial Domain Adaptation (ViT-ADA) (Li & Wu, 2024) represents a recent effort that integrates adversarial domain adaptation directly into Vision Transformer representations to mitigate heterogeneous scanner-induced shifts. Dual Cross-image Semantic Consistency (DCSC) (Wu et al., 2025) is another state-of-the-art approach that enforces semantic consistency across images to improve robustness under complex cross-site discrepancies. Both methods represent the latest advances in heterogeneous MRI adaptation and serve as strong modern baselines.

5.2. Results of binary classification

Table 3 reports the comparative performance of multiple domain adaptation approaches on the binary classification task distinguishing cognitively normal individuals with and without depression (CN-D vs. CN-N). We also report the p -values from paired sample t -test to assess whether the performance differences between the proposed CDA and specific competing methods are statistically significant (threshold: $p < .05$). From Table 3, we have the following observations. *First*, traditional unsupervised domain adaptation (UDA) methods based on handcrafted MRI features, such as TCA, SCA, and JDA, show substantially inferior performance (AUC: 49.62%-54.15%, ACC: 50.75%-54.35%) when compared to the end-to-end deep learning models. This performance gap likely stems from the limited capacity of handcrafted features to capture the complex neuroanatomical variations present in 3D structural MRI. In contrast, deep models operating directly on raw MRI volumes achieve markedly better results, highlighting the superiority of learned hierarchical representations for LLD detection. *Second*, among MMD-based deep learning approaches, DAN and DSAN demonstrate effective alignment via multi-layer and subdomain-level feature adaptation. However, their performance remains consistently inferior to that of CDA, highlighting the added benefits of collaborative and bidirectional adaptation strategies. *Third*, adversarial learning-based methods such as DAAN also performed well in terms of AUC and ACC, indicating a relatively strong capacity to handle domain shifts. However, other adversarial approaches like ADDA lagged significantly behind across nearly all metrics, especially in terms of SPE and F1, suggesting unstable performance and limited generalization to the target domain. *Fourth*, information-theoretic models like BNM and SHOT++ aim to improve class separability. While BNM achieves competitive performance among deep UDA methods, SHOT++ lags behind, highlighting substantial differences in their generalization capabilities. *Finally*, the CDA framework achieves the highest performance across all metrics, outperforming all competing methods in this task. Statistical significance testing further confirms that CDA's improvements are significant ($p < .05$) compared to most competing methods, suggesting the robustness and reliability of CDA in addressing domain shift challenges in cross-site MRI analysis.

5.3. Results of three-Category classification

For CI vs. CN-D vs. CN-N classification, we report the ACC and SEN results achieved by different methods for each category in Tables 4 and 5, respectively. As shown in Table 4, our CDA method achieves the highest overall ACC result as well as the highest ACC value for the

Table 3

Results achieved by different domain adaptation methods (mean $\pm$ standard deviation) for CN-D vs. CN-N classification.

Method	AUC (%)	ACC (%)	SEN (%)	SPE (%)	F1 (%)	<i>p</i> -value
TCA	49.62 $\pm$ 1.95	50.75 $\pm$ 1.70	63.88 $\pm$ 6.13	35.29 $\pm$ 8.32	39.18 $\pm$ 6.53	0.0001
JDA	54.15 $\pm$ 1.38	54.35 $\pm$ 1.12	62.03 $\pm$ 1.45	46.27 $\pm$ 4.01	50.54 $\pm$ 4.61	0.0002
SCA	51.95 $\pm$ 2.76	52.10 $\pm$ 1.66	58.26 $\pm$ 8.26	37.84 $\pm$ 17.14	32.11 $\pm$ 4.45	0.0005
DAN	68.87 $\pm$ 0.38	62.00 $\pm$ 3.08	67.15 $\pm$ 3.14	50.00 $\pm$ 13.69	71.23 $\pm$ 1.69	0.0012
DAAN	67.37 $\pm$ 10.98	65.28 $\pm$ 3.38	68.89 $\pm$ 7.54	42.73 $\pm$ 21.97	55.15 $\pm$ 16.13	0.0327
DeepCORAL	68.87 $\pm$ 0.65	62.00 $\pm$ 1.00	67.86 $\pm$ 3.09	48.33 $\pm$ 8.07	71.40 $\pm$ 1.07	0.0079
ADDA	52.39 $\pm$ 3.64	51.53 $\pm$ 13.94	64.45 $\pm$ 10.08	36.07 $\pm$ 12.35	53.78 $\pm$ 9.80	0.0002
DANN	57.88 $\pm$ 9.46	57.93 $\pm$ 4.58	66.57 $\pm$ 7.43	46.16 $\pm$ 10.15	47.49 $\pm$ 8.60	0.0003
BNM	68.87 $\pm$ 0.78	60.00 $\pm$ 3.06	67.15 $\pm$ 15.42	43.33 $\pm$ 17.10	71.14 $\pm$ 1.59	0.0063
DSAN	68.81 $\pm$ 0.25	61.00 $\pm$ 2.92	68.57 $\pm$ 3.61	43.33 $\pm$ 3.73	71.07 $\pm$ 2.40	0.0079
SHOT + +	54.00 $\pm$ 10.13	50.37 $\pm$ 8.21	54.76 $\pm$ 12.27	56.67 $\pm$ 10.16	52.37 $\pm$ 10.57	0.0056
ViT-ADA	70.60 $\pm$ 5.80	68.40 $\pm$ 8.90	68.50 $\pm$ 7.20	74.50$\pm$11.00	70.20 $\pm$ 9.50	0.0545
DCSC	68.20 $\pm$ 6.50	64.80 $\pm$ 9.20	72.30 $\pm$ 7.50	62.50 $\pm$ 10.00	63.50 $\pm$ 9.80	0.0523
CDA (Ours)	71.51$\pm$6.82	70.73$\pm$8.59	69.29$\pm$8.32	73.09 $\pm$ 12.42	74.99$\pm$10.62	–

Table 4

Accuracy (ACC) results achieved by different methods (mean $\pm$ standard deviation) in the task of three-category classification (i.e., CI vs. CN-D vs. CN-N) on the target data.

Method	ACC (%)	ACC _{CN-N} (%)	ACC _{CN-D} (%)	ACC _{CI} (%)	<i>p</i> -value
TCA	40.81 $\pm$ 7.21	57.98 $\pm$ 6.91	45.82 $\pm$ 8.65	67.83 $\pm$ 7.28	0.0007
JDA	44.22 $\pm$ 5.57	60.97 $\pm$ 3.09	52.71 $\pm$ 8.20	74.76 $\pm$ 7.21	0.0006
SCA	45.09 $\pm$ 13.90	69.66$\pm$5.43	59.87 $\pm$ 9.67	72.92 $\pm$ 8.07	0.0180
DAN	34.59 $\pm$ 17.44	60.73 $\pm$ 16.56	49.33 $\pm$ 9.41	59.12 $\pm$ 12.65	0.0023
DeepCORAL	36.17 $\pm$ 6.96	59.72 $\pm$ 7.10	43.69 $\pm$ 2.18	68.93 $\pm$ 11.42	0.0004
DSAN	43.33 $\pm$ 7.30	61.33 $\pm$ 16.14	52.67 $\pm$ 2.49	72.67 $\pm$ 8.54	0.0007
DANN	48.83 $\pm$ 11.71	62.19 $\pm$ 9.34	54.84 $\pm$ 7.59	80.64 $\pm$ 9.55	0.0020
ADDA	44.13 $\pm$ 16.89	63.77 $\pm$ 15.98	52.33 $\pm$ 6.47	72.17 $\pm$ 17.78	0.0362
DAAN	50.09 $\pm$ 5.86	68.57 $\pm$ 4.88	56.54 $\pm$ 4.44	80.70 $\pm$ 2.75	0.0449
BNM	34.00 $\pm$ 4.52	67.33 $\pm$ 4.42	48.00 $\pm$ 5.42	48.67 $\pm$ 9.57	0.0000
SHOT + +	45.57 $\pm$ 7.87	53.87 $\pm$ 13.38	49.43 $\pm$ 16.63	82.58$\pm$3.57	0.0039
ViT-ADA	50.60 $\pm$ 10.08	60.67 $\pm$ 14.43	55.26 $\pm$ 21.68	47.88 $\pm$ 6.64	0.4556
DCSC	46.45 $\pm$ 8.57	68.64 $\pm$ 15.33	55.07 $\pm$ 2.53	69.19 $\pm$ 3.66	0.0497
CDA (Ours)	50.79$\pm$8.93	59.49 $\pm$ 6.75	64.22$\pm$5.32	64.50 $\pm$ 10.00	–

CN-D category, outperforming all competing models. CDA also shows more balanced performance across the three categories, with ACC values of 59.49% for CN-N, 64.22% for CN-D, and 64.50% for CI. The three deep learning methods (DANN, DAAN, and SHOT + +) yield very imbalanced results among the three categories, with lower overall accuracy compared with CDA. The results in Table 5 suggest that our CDA obtains the highest overall SEN and the highest SEN for CN-N and comparable performance for CN-D, but the sensitivity for CI is not good. Notably, several methods, such as DANN, DAAN, and DSAN, obtain relatively high SEN scores for CN-D, but their performance on CI and CN-N is substantially lower, which again reflects class imbalance in predictions. Although CDA achieves reasonable SEN for CN-D and CN-N, the SEN value for the CI class remains suboptimal. This may be due to the limited number of CI subjects (only 17), and the even smaller representation of this class in each cross-validation test fold. Such data imbalance likely hinders the model's ability to generalize effectively in multi-category classification.

Comparing the three-category classification results in Tables 4-5 with the binary classification results in Table 3, we observe a notable decline in overall performance when transitioning from binary to multi-class classification. This performance drop is largely attributable to the increased difficulty in accurately identifying the newly introduced CI category. The low sensitivity for the CI class observed on NBOLD could be interpreted in the context of extreme data scarcity. More evaluation on a large target dataset can be found in Section 6.9.

6. Discussion

6.1. Ablation study

To evaluate the contribution of each stage in our CDA framework, we compare CDA with four ablated variants: CDA_{S₁}, CDA_{S₁₂}, CDA_{S₁₃}, and CDA_{S₂₃}. Specifically, CDA_{S₁} involves only Stage 1 (supervised training on labeled source data). CDA_{S₁₂} incorporates both Stage 1 and Stage 2 (self-supervised target feature adaptation). CDA_{S₁₃} combines Stage 1 with Stage 3 (collaborative training on unlabeled target data). CDA_{S₂₃} utilizes only Stage 2 and Stage 3, without supervised source training. All variants use the same backbone and hyperparameters to ensure a fair comparison.

As shown in Fig. 2, the CDA achieves the overall best performance. For instance, CDA yields an AUC of 71.51%, which is higher than all baselines. Similarly, CDA obtains the highest F1 of 74.99%, outperforming its variants by 1.90%-5.70%, demonstrating its advantage in handling class imbalance and improving discriminative capacity. Notably, both CDA_{S₁₂} and CDA_{S₁₃} outperform CDA_{S₁} in most cases, demonstrating that incorporating either self-supervised adaptation (Stage 2) or collaborative training (Stage 3) leads to improved generalization over supervised learning alone. The full CDA framework, which integrates all three stages, achieves the most substantial performance gains, validating the complementary benefits of these strategies. Moreover, CDA_{S₂₃} also shows competitive results, even without supervised source

Table 5

Sensitivity (SEN) results achieved by different methods (mean $\pm$ standard deviation) in the task of three-category classification (i.e., CI vs. CN-D vs. CN-N) on the target data.

Method	SEN (%)	SEN _{CN-N} (%)	SEN _{CN-D} (%)	SEN _{CI} (%)	p-value
TCA	34.22 $\pm$ 2.72	21.43 $\pm$ 11.19	59.58 $\pm$ 10.73	21.67 $\pm$ 9.44	0.0430
JDA	37.64 $\pm$ 7.88	39.64 $\pm$ 19.90	53.28 $\pm$ 1.88	20.00 $\pm$ 1.87	0.0887
SCA	33.81 $\pm$ 0.95	32.50 $\pm$ 20.22	63.29 $\pm$ 24.95	43.33$\pm$16.16	0.0194
DAN	35.24 $\pm$ 2.33	28.57 $\pm$ 17.25	37.14 $\pm$ 15.71	40.00 $\pm$ 8.99	0.0375
DeepCORAL	31.28 $\pm$ 7.69	21.07 $\pm$ 11.32	46.10 $\pm$ 21.97	26.67 $\pm$ 7.59	0.0418
DSAN	31.11 $\pm$ 1.88	20.00 $\pm$ 21.79	70.00 $\pm$ 23.18	3.33 $\pm$ 6.67	0.0084
DANN	37.95 $\pm$ 11.14	22.86 $\pm$ 11.03	69.32 $\pm$ 14.33	21.67 $\pm$ 9.63	0.0487
ADDA	33.33 $\pm$ 0.34	20.00 $\pm$ 4.89	60.00 $\pm$ 4.53	20.00 $\pm$ 2.67	0.0041
DAAN	35.41 $\pm$ 3.77	25.36 $\pm$ 6.59	70.88$\pm$14.93	10.00 $\pm$ 2.00	0.0588
BNM	29.58 $\pm$ 4.90	7.50 $\pm$ 10.00	41.25 $\pm$ 14.03	40.00 $\pm$ 22.61	0.0078
SHOT + +	34.20 $\pm$ 5.31	42.86 $\pm$ 15.75	59.74 $\pm$ 12.14	6.67 $\pm$ 13.33	0.0189
ViT-ADA	38.33 $\pm$ 6.19	33.33 $\pm$ 22.16	58.33 $\pm$ 20.14	23.27 $\pm$ 28.87	0.0472
DCSC	34.01 $\pm$ 3.56	21.67 $\pm$ 22.32	68.46 $\pm$ 24.08	11.90 $\pm$ 16.77	0.0361
CDA (Ours)	42.22$\pm$6.52	50.67$\pm$3.79	63.33 $\pm$ 4.52	12.67 $\pm$ 1.33	–

Table 6

Quantitative comparison of classification performance (accuracy and class-wise sensitivity) on the ADNI (source) $\rightarrow$ AIBL (target) benchmark. Results are reported as “mean $\pm$ std”. Best results are shown in bold.

Method	SEN _{CN} (%)	SEN _{MCI} (%)	SEN _{AD} (%)	Macro SEN (%)	Total ACC (%)	p-value (SEN)
SHOT + +	40.00 $\pm$ 18.99	23.87 $\pm$ 27.69	46.67 $\pm$ 15.22	33.50 $\pm$ 0.34	57.97 $\pm$ 11.72	0.0018
ViT-ADA	66.48$\pm$21.13	36.03 $\pm$ 17.08	36.10 $\pm$ 29.22	42.87 $\pm$ 4.99	66.71 $\pm$ 8.80	0.0395
DCSC	52.29 $\pm$ 12.46	38.33 $\pm$ 8.72	53.86 $\pm$ 13.42	45.16 $\pm$ 9.27	63.36 $\pm$ 11.23	0.0435
CDA (Ours)	47.14 $\pm$ 8.99	55.13$\pm$17.25	54.29$\pm$14.86	48.85$\pm$4.72	68.04$\pm$5.49	–

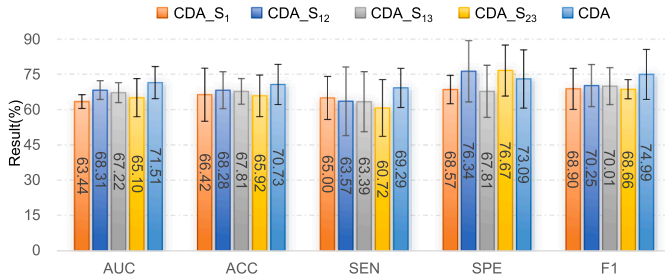


Fig. 2. Ablation study comparing CDA and its four variants in the task of CN-D vs. CN-N classification.

training. In particular, it consistently outperforms the source-only baseline (CDA_{S1}) across key evaluation metrics (e.g., +4.28% in ACC), indicating that the proposed collaborative learning strategy can effectively exploit target-domain structure even without source supervision. This further emphasizes the effectiveness of target-domain self-supervision and cross-branch collaboration. These findings confirm the efficacy of CDA’s staged components for LLD identification.

6.2. Effect of dual-Branch backbone architecture

To investigate the effect of different backbone architectures on performance, we conduct a comparative study across three configurations: CDA_{CNN+CNN} with two CNN branches, CDA_{ViT+ViT} with two ViT branches, and our proposed hybrid ViT + CNN used in CDA. The experimental results are presented in Fig. 3. As shown in this figure, the hybrid CDA model consistently outperforms both homogeneous variants across all evaluation metrics. Specifically, it achieves the highest AUC, ACC, SEN, SPE, and F1, significantly surpassing both CDA_{CNN+CNN}, CDA_{ViT+ViT}. These results suggest the benefit of integrating global and local representations within a unified framework. The ViT branch contributes strong global context modeling, while the CNN branch captures fine-grained local texture information. This combination enhances feature diversity and leads to more effective cross-domain alignment. Over-

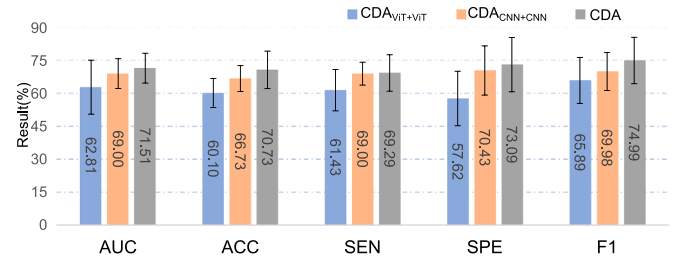


Fig. 3. Performance comparison of CDA and its two variants (i.e., CDA_{ViT+ViT} and CDA_{CNN+CNN}) employing different backbone architectures for CN-D vs. CN-N classification.

all, the empirical evidence validates the effectiveness of the hybrid ViT-CNN architecture in CDA, demonstrating its superiority over purely homogeneous designs.

6.3. Hyperparameter analysis

For collaborative training on the unlabeled target data, the selection of threshold values in Eqs. (7)–(8) plays a crucial role in determining the overall training effectiveness. Fig. 4 shows the impact of varying thresholds θ_2 and θ_1 on model performance. In Fig. 4 (a), we fix $\theta_1 = 0.5$ and vary θ_2 from 0.5 to 0.9. As θ_2 increases, both AUC and ACC improve steadily, peaking at $\theta_2 = 0.8$, and slightly declining at 0.9, suggesting that the model performs robustly near $\theta_2 = 0.8$.

In Fig. 4 (b), we investigate the sensitivity of θ_1 while fixing $\theta_2 = 0.8$ (the optimal value identified above). The results indicate that the ViT branch is generally robust to threshold variations. We observed that setting $\theta_1 = 0.5$ achieves optimal or highly competitive performance. Unlike the CNN branch, a moderate threshold for ViT (e.g., 0.5) is sufficient to filter noise while preserving the diversity of global semantic guidance. Higher thresholds for θ_1 did not yield significant gains and potentially limited the scope of collaborative supervision. Therefore, we adopted $\theta_1 = 0.5$ and $\theta_2 = 0.8$ as the final configuration to balance pseudo-label quality and quantity.

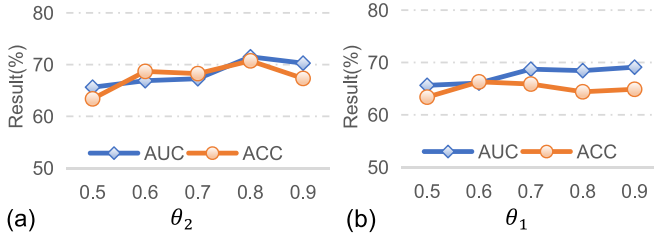


Fig. 4. Results of CDA with different threshold combinations in the task of CN-D vs. CN-N classification.

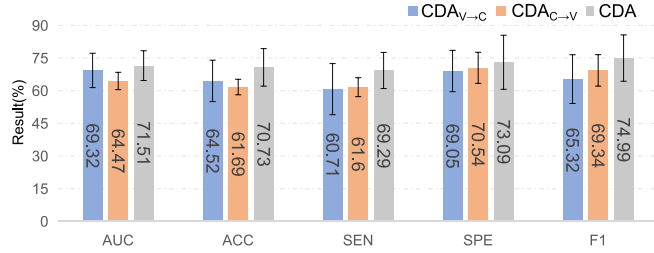


Fig. 5. Comparison of CDA and its two variants with different collaborative training strategies for CN-D vs. CN-N classification.

6.4. Influence of collaborative training strategy

To validate the benefits of collaborative learning in CDA, we conduct an ablation study comparing CDA with two variants that use different collaborative training strategies: $CDA_{V \rightarrow C}$, and $CDA_{C \rightarrow V}$. Both variants share the same first two stages as the original CDA framework. In the third stage of $CDA_{V \rightarrow C}$, the ViT branch generates pseudo labels from its predictions on unlabeled target data to guide the CNN branch by minimizing the loss $\mathcal{L}_{V \rightarrow C}$. Conversely, in $CDA_{C \rightarrow V}$, the CNN branch generates pseudo labels to supervise the ViT branch by minimizing $\mathcal{L}_{C \rightarrow V}$. Results in Fig. 5 show that the full CDA framework achieves the highest results across all metrics. For instance, $CDA_{V \rightarrow C}$ achieves 69.32% AUC and 64.52% ACC, indicating limited adaptation when only the ViT branch provides supervision. $CDA_{C \rightarrow V}$ performs even worse, showing further declines in both metrics. Notably, $CDA_{V \rightarrow C}$ still underperforms the full CDA by 2.19% in AUC, indicating the complementary contributions of both modality-specific branches and the advantage of their integration. These observations demonstrate that one-directional supervision is suboptimal due to the asymmetric nature and learning capacity of CNN and ViT architectures. In contrast, the fully cooperative training strategy-wherein the two branches mutually exchange pseudo-labels and iteratively refine their predictions-demonstrated superior performance across all evaluation metrics. This significant gain validates that the bidirectional pseudo-labeling encourages mutual enhancement and alignment between the global (ViT) and local (CNN) feature representations. The collaborative learning strategy plays a critical role in enhancing domain adaptation by balancing representation diversity and decision consistency.

6.5. Effect of inference model

While the main experiments utilize the trained CNN model for inference in CDA, we also evaluate a variant employing the ViT model for inference, referred to as **CDA-V**. The results, reported in Fig. 6, show that CDA-V achieves 67.30% AUC, 65.08% ACC, 63.93% SEN, 72.98% SPE, and 66.72% F1, all lower than CDA's corresponding metrics. The experimental results suggest that CNN-based local representations may provide superior discriminative capacity compared to ViT's global modeling when applied to target domain inference. Although ViT is known for capturing global context, its standalone performance appears limited

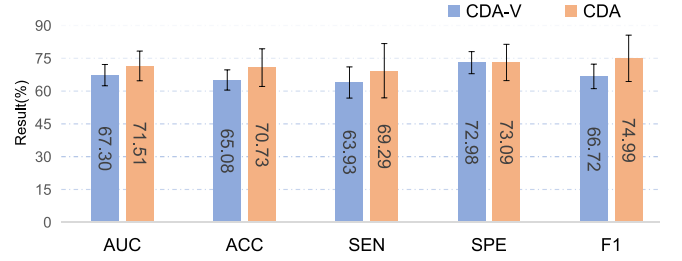


Fig. 6. Comparison between CDA (with CNN for inference) and CDA-V (with ViT for inference) in the task of CN-D vs. CN-N classification.

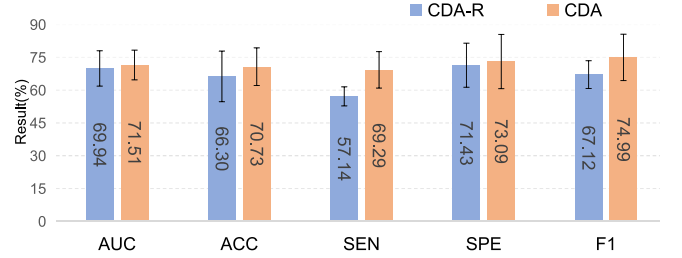


Fig. 7. Comparison between CDA and its variant CDA-R with different encoder fine-tuning strategy for CN-D vs. CN-N classification.

in this setting, potentially due to a lower inductive bias and greater sensitivity to limited data. In contrast, CNN strengthened through collaborative training and pseudo-label refinement effectively captures local features while benefiting from the stability introduced by joint supervision during collaborative training. These findings strongly support the core hypothesis of CDA: the collaborative training framework enables the CNN to harness ViT's generalization strengths while preserving robust and precise decision boundaries through local receptive fields, thus improving cross-domain classification.

6.6. Influence of encoder fine-tuning

To assess the necessity of asymmetric design in Stage 2 for target-source feature alignment (see Fig. 1 (b)), we carry out a controlled experiment by interchanging the roles of ViT and CNN branches during discrepancy optimization. In the original CDA setting, we freeze the ViT encoder (E_V) while updating the two classifiers to maximize their discrepancy. We then fine-tune the CNN encoder (E_C) while keeping the classifiers fixed, minimizing their discrepancy to achieve stable feature alignment. To evaluate the sensitivity of this design, we reverse the roles of the two branches: we first freeze the CNN encoder while updating the classifiers, and then fine-tune the ViT encoder with the classifiers frozen. We refer to this variant as **CDA-R**. The resulting performance of CDA and CDA-R is reported in Fig. 7. As shown, CDA-R exhibits a notable decline compared to CDA, particularly in SEN, which drops to 57.14%. This implies that CNN lacks the global capacity needed for effective discrepancy maximization, while ViT is more sensitive to noisy pseudo-labels during alignment. These results emphasize the importance of CDA's asymmetric design: the ViT encoder promotes boundary diversification, while the CNN encoder ensures stable alignment, together enabling effective cross-domain adaptation.

6.7. Influence of source domain

We use NCODE as the source domain in the main experiments. To further assess the adaptability of our CDA with different source domains, we perform a comparative study using ADNI (Jack et al., 2008) as the source domain, while maintaining NBOLD as the target domain.

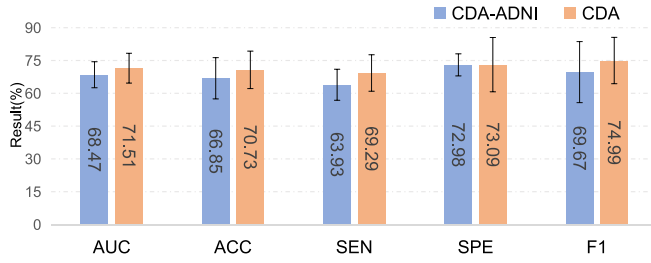


Fig. 8. Results of our CDA (with NCODE as the source domain) and its variant CDA-ADNI (with ADNI as the source domain) for CN-D vs. CN-N classification on the target NBOLD domain.

We refer to this method as **CDA-ADNI**. For CDA-ADNI, the baseline T1-weighted MRI scans from CN and AD groups in ADNI-1 and ADNI-2 are used as source data, including 359 AD and 436 CN subjects.

As summarized in Fig. 8, training CDA with NCODE as the source domain consistently outperforms its counterpart trained on ADNI (CDA-ADNI) across all evaluation metrics. Specifically, CDA achieves improvements of +3.04% in AUC, +3.88% in ACC, +5.36% in SEN, +0.11% in SPE, and +5.32% in F1 relative to CDA-ADNI. This performance gap is largely attributable to the greater domain similarity between NCODE and the target NBOLD dataset, including more consistent imaging protocols, better demographic alignment, and a shared diagnostic taxonomy. Despite this domain gap, CDA-ADNI still delivers competitive results, owing to the substantial volume of training data it provides. Notably, the ADNI dataset includes 795 MRI scans, offering rich structural variability and a broader representation space that serves as strong auxiliary supervision for downstream tasks. These findings highlight a key insight: while domain proximity facilitates feature transferability and decision boundary alignment, large-scale auxiliary datasets enhance representation robustness and mitigate overfitting to domain-specific artifacts. The proposed CDA framework adapts well to different source data conditions. It takes advantage of semantic similarity when available (as with NCODE) and benefits from large, diverse datasets when similarity is lower (as with ADNI), showing strong flexibility in cross-domain neuroimaging classification.

6.8. Impact of encoder pretraining

In the main experiments, we employ pretrained encoders (E_V and E_C) in ViT and CNN, respectively. Specifically, E_V is pretrained on a combination of three datasets (IXI, OASIS-3, and BRATS) (Kunabayeve et al., 2024), while E_C is pretrained on ADNI (Zhang et al., 2025). To assess the impact of this encoder pretraining strategy, we compare our CDA with a baseline variant, **CDAw/oP**, which uses randomly initialized encoders without any pretraining. The results of the two methods are reported in Fig. 9. As illustrated in this figure, CDA with encoder pretraining consistently outperforms CDAw/oP on downstream tasks: AUC increases by 2.67%, ACC by 3.52%, SEN by 8.57%, SPE by 4.04%, and F1 score by 5.65%. These performance gains validate the effectiveness of our pretraining strategy, which leverages structural prior knowledge to learn more transferable and domain-invariant representations. The findings are consistent with prior work (Zhang et al., 2025), while further extending its applicability to domain adaptation scenarios. By encoding fundamental anatomical patterns in MRI during pretraining, the model gains robustness against inter-domain variability and becomes less dependent on large labeled samples in the source domain.

6.9. Validation on additional dataset

To further investigate the model's sensitivity and generalization capability under less imbalanced data settings, we conducted an additional evaluation on the AIBL (Ellis et al., 2009) dataset (target domain) us-

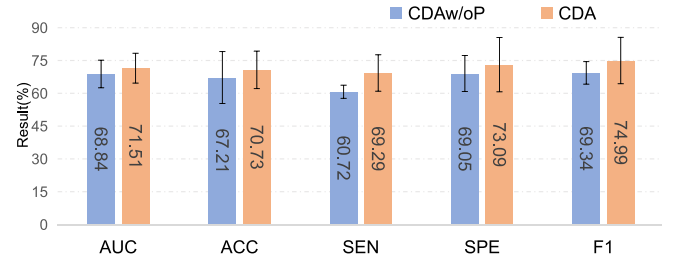


Fig. 9. Comparison between CDA and its variant (CDAw/oP) without encoder pretraining for CN-D vs. CN-N classification.

ing ADNI as the source domain. AIBL contains 447 CN, 63 mild cognitive impairment (MCI), and 71 AD subjects, while ADNI contains 436 CN, 291 MCI, and 359 AD subjects. The corresponding results are reported in Table 6. Unlike the NBOLD target domain, where the minority cognitive impairment (CI) class is extremely scarce ($N = 17$), the AIBL dataset provides a substantially larger and more representative number of samples for each category. As observed, CDA achieves a significantly improved sensitivity of 55.13% for MCI and 54.29% for AD, consistently outperforming state-of-the-art baselines like ViT-ADA and DCSC. This implies that the low sensitivity observed in the main experiments is driven by the scarcity of minority-category target data. When provided with a representative test distribution, our class-balanced learning strategy effectively captures minority class features.

6.10. Limitations and future work

Although our CDA demonstrates strong performance, several limitations remain to be addressed. *First*, the current model relies on a fixed backbone architecture (ViT and CNN) without further architectural optimization for the LLD detection task. Future work could explore the integration of more advanced backbones, such as hybrid transformer-convolutional architectures or domain-specific neural operators, to better capture complex neuroanatomical patterns in LLD cohorts. *Second*, while CDA achieves robust cross-domain performance, its generalization to entirely unseen imaging sites with more extreme distribution shifts remains to be evaluated. In future studies, incorporating domain generalization strategies or meta-learning paradigms may further enhance adaptability. *Lastly*, multimodal neuroimaging data (e.g., functional MRI, DTI) could provide additional complementary information and potentially further improve diagnostic accuracy. However, this study intentionally focuses on T1-weighted structural MRI, which is the most widely available and cost-effective modality in routine clinical and geriatric settings, whereas functional and diffusion imaging are often unavailable or incomplete in real-world scenarios. Extending CDA to multi-modal neuroimaging data remains an important direction for future work.

7. Conclusion

This study introduces a CDA framework for automated LLD identification using T1-weighted MRI scans. CDA combines a ViT and a CNN to extract global and local brain features. Its training consists of three stages: supervised learning on source data, self-supervised adaptation to refine decision boundaries, and collaborative training on unlabeled target data using confidence-filtered pseudo-labels. Comprehensive experiments on multi-site datasets show that CDA generally outperforms state-of-the-art methods in two classification tasks. These results highlight its strong generalizability under domain shifts, making it a promising tool for real-world LLD detection.

CRedit authorship contribution statement

Yuzhen Gao: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis; **Qianqian Wang:** Writing – review & editing; **Yongheng Sun:** Writing – review & editing, Software, Methodology; **Cui Wang:** Writing – review & editing; **Yongquan Liang:** Writing – review & editing; **Mingxia Liu:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found in the online version at [10.1016/j.neunet.2026.108911](https://doi.org/10.1016/j.neunet.2026.108911).

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